

Week 2, Lecture 2: Bestiary of Functions

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker 3.5 (“Bestiary of functions”), leaning heavily on Table 3.1 (p. 119) and section 3.5.2 (exponential-based functions, p. 127-130) for the logistic. Also mined: this morning’s WOD (`wod/WOD_Week2_Lecture2_BestiaryLimits.Rmd`), Monday’s lecture notes, and Lab 2 Acts 1-2 (flexible Ricker, Holling IV).

Where this sits in the week

- Monday: the toolkit – limits (throw out lower-order terms), derivatives, critical points – worked through Michaelis-Menten and Ricker.
- Monday’s WOD: R syntax + derived/coded a seasonal sine function whose steepest point (day 91) was never named.
- Tuesday (Lab 2): Act 1 (grid search), Act 2 (flexible Ricker, Holling IV – another rational function, one degree harder than Michaelis-Menten).
- Wednesday AM (WOD4, already done): limits practice on four functions (two familiar – Michaelis-Menten, Ricker – and two new – logistic, Holling III), then a derivative of the logistic and a leading question about what kind of point $x = 0$ is.
- **Wednesday lecture (this one):** reveal WOD4, name the new concept it was building toward (inflection points), and give the map of the bestiary that today’s and Monday’s examples are drawn from.

Rationale for the flow

WOD4’s Part 2 is a setup, not a self-contained exercise – it walks students to the edge of “the logistic’s midpoint has a nonzero derivative, so it’s not a critical point... so what is it?” and leaves the actual concept (inflection point, concavity, second derivative) unnamed, with only a hint. That’s today’s centerpiece: give the reveal, not a whole new derivation from scratch. The payoff is bigger than just the logistic – Monday’s WOD sine function has the *exact same* feature at day 91 (maximum slope, not zero slope), and nobody named it then either. Connecting those two moments is worth building the lecture around.

Bolker himself says to treat 3.5 as a map, not a memorization list: “*You should skim through*

this section on the first reading to get an idea of what functions are available... use the section for reference as needed. So the bestiary-survey portion of this lecture is deliberately shallow and wide – a table, not a derivation marathon – with one deeper worked example (Holling III’s limits, which double-checks WOD4 and sets up a “staircase” of difficulty: MM’s numerator and denominator have different degrees; Holling III’s have the *same* degree; Lab 2’s Holling IV has the same degree plus an extra middle term. Today is the middle rung.

Part A: reveal WOD4’s limits (quick – ~5 min)

1a-1b (Michaelis-Menten, Ricker): already fully worked Monday – just flash the answers, don’t re-derive.

- MM: $x \rightarrow 0$ gives a line through the origin, slope a/b ; $x \rightarrow \infty$ gives the asymptote a .
- Ricker: $x \rightarrow 0$ gives 0 (both factors vanish); $x \rightarrow \infty$ gives 0 (exponential beats the linear x term).

1c (Logistic), $f(x) = \frac{1}{1 + e^{-bx}}$: as $x \rightarrow \infty$, $e^{-bx} \rightarrow 0$, so $f(x) \rightarrow 1/(1+0) = 1$. As $x \rightarrow -\infty$ (or, for the WOD’s $x \rightarrow 0$ framing, just evaluate directly), $e^{-bx} \rightarrow \infty$ so $f(x) \rightarrow 1/\infty = 0$. The logistic is bounded between 0 and 1 – no new dominance trick needed, just careful bookkeeping of what e^{-bx} does at each end.

1d (Holling III), $f(x) = \frac{ax^2}{b^2 + x^2}$ – **the one worth doing on the board:**

- $x \rightarrow 0$: numerator $ax^2 \rightarrow 0$ (it’s the higher-order term here, since there’s no linear term to compete with); denominator $b^2 + x^2 \approx b^2$ (constant wins). So $f(x) \approx ax^2/b^2 \rightarrow 0$, and *quadratically* – the curve leaves the origin flat (slope zero) and curves upward, unlike Michaelis-Menten’s straight-line start.
- $x \rightarrow \infty$: **numerator and denominator are now the *same* degree** (x^2 on both sides) – new wrinkle beyond Monday. Throw out the lower-order term in the denominator ($b^2 \ll x^2$), leaving $f(x) \approx ax^2/x^2 = a$. Same asymptote as Michaelis-Menten, same logic, just both leading terms survive and cancel instead of one side disappearing entirely.

Sticking point to flag: when numerator and denominator are the same degree, students sometimes think the dominance trick “doesn’t apply” because nothing fully vanishes. It still applies – you throw out the lower-order term *within* each of the numerator and denominator separately, then simplify what’s left. This is exactly the skill Lab 2 Act 2 needs for Holling IV (same idea, with an extra cx term to also discard in the denominator).

Part B: critical points vs. inflection points (the core new idea, ~12-14 min)

Recap the vocabulary from Monday first: a critical point is where $f'(x) = 0$ – a candidate peak or trough. Ricker has one; Michaelis-Menten, being monotonic, has none.

Now work the logistic's derivative on the board (this is WOD4 2a-2b, revealed): $f(x) = (1 + e^{-bx})^{-1}$. Chain rule, twice:

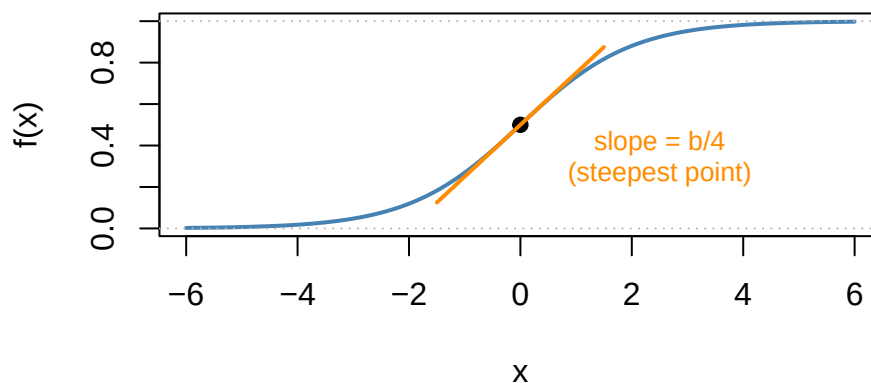
$$f'(x) = -(1 + e^{-bx})^{-2} \cdot (-b e^{-bx}) = \frac{b e^{-bx}}{(1 + e^{-bx})^2}$$

```
b <- 1
f <- function(x) 1 / (1 + exp(-b * x))
fp <- function(x) b * exp(-b * x) / (1 + exp(-b * x))^2
fp(0)
```

```
## [1] 0.25
```

At $x = 0$: $e^0 = 1$, so $f'(0) = b \cdot 1/(1 + 1)^2 = b/4$ – **not zero** (for any $b \neq 0$). This matches Bolker's own Figure 3.9, which labels exactly this point “ $b/4$ ” on the logistic panel.

So $x = 0$ is not a critical point – what is it? It's the point of *steepest* slope on the whole curve: the logistic climbs slowly, then fastest right around $x = 0$, then slowly again as it flattens toward the asymptote at 1. A point where the slope is at a local maximum (rather than zero) is an **inflection point**: formally, where the *second* derivative $f''(x) = 0$ and changes sign – where the curve's concavity flips from curving upward (concave up) to curving downward (concave down). Don't grind through $f''(x)$ symbolically today (that's more than WOD4 asked for); the geometric argument is enough: just before $x = 0$ the curve bends upward (accelerating), just after it bends downward (decelerating) – concavity changes, so it's an inflection point, not a critical point.



The payoff – callback to Monday’s WOD: the seasonal sine function students derived Monday (period 1 year, steepest at day 91) has exactly this feature. Day 91 wasn’t a critical point either – the function wasn’t at a max or min there, it was climbing at its *fastest* rate. Same concept, unnamed until right now. Worth saying explicitly: *every sigmoid has exactly one inflection point (its steepest point); every periodic function like the sine has one every half-cycle*. Two completely different-looking functions, same calculus feature.

Sticking point to flag (restated from Monday, now with a second example): critical point ($f' = 0$, zero slope) vs. inflection point (steepest slope, $f'' = 0$ with a sign change) are easy to blur together because both involve “a special point found using a derivative.” Keep the plain-English distinction front and center: critical point = *flattest* nearby; inflection point = *steepest* nearby (for a sigmoid; more generally, just “concavity changes here”).

Part C: the bestiary map (~8-10 min)

Bolker’s Table 3.1 organizes the whole bestiary into four families. Don’t derive every row – this is a map, students already own several of these landmarks:

- **Polynomial / piecewise** – line, quadratic, cubic; threshold, hockey stick. *Not yet met.*
- **Rational (ratios of polynomials)** – hyperbolic, Michaelis-Menten, Holling III, Holling IV. *Already met: MM (Monday), Holling IV (Lab 2), Holling III (this morning’s WOD).*
- **Exponential-based** – negative exponential, monomolecular, **Ricker**, **logistic**. *Already met: Ricker (Monday, Lab 1-2), logistic (this morning’s WOD).*
- **Power-based** – power law, von Bertalanffy, Gompertz, Shepherd. *Not yet met.*

Point out explicitly: **you have already personally met roughly half this table** without necessarily realizing it was “the bestiary” – today just gives it a name and a map.

One mechanism worth telling, since it’s the most famous equation in ecology: the logistic function is the *solution* to the logistic growth equation,

$$\frac{dn}{dt} = rn \left(1 - \frac{n}{K}\right)$$

– per-capita growth rate declines linearly from r (when $n \ll K$) to zero (when $n = K$, the carrying capacity). The equation is separable; the resulting integral $\int \frac{dn}{n(1-n/K)}$ is solved via *partial fraction decomposition* ($\frac{K}{n(K-n)} = \frac{1}{n} + \frac{1}{K-n}$), not shown today, and produces exactly the sigmoid curve just derived. This is a nice moment to connect the purely-curve-fitting view of “logistic = an S-shaped function that happens to fit data” with the mechanistic view every student already has from intro ecology: population growth that decelerates as it approaches a ceiling.

Briefly mention, don’t derive: the Gompertz function is an alternative sigmoid whose inflection point sits at $1/e \approx 1/3$ of the way up rather than halfway – worth one sentence as a reminder that “sigmoid” is a shape category with more than one member, the same way “hump-shaped” wasn’t just Ricker (flexible Ricker, Holling IV).

Take-home message

- Two new limit moves today: same-degree rational functions (Holling III/IV) still use dominance, just applied separately to numerator and denominator before taking the ratio.
- Inflection point $\neq$ critical point: steepest slope, not zero slope; concavity change, not a peak/trough.
- The logistic's midpoint and the sine's day-91 point (Monday's WOD) are the same calculus feature wearing two different costumes.
- The bestiary is a map, not a memorization list – four families (polynomial, rational, exponential-based, power-based), and students already own examples from three of the four.

Roadmap tie-in

- Week 3: probability and distributions – deterministic functions (this week) become the *mean* or *expected* part of a full stochastic model; today's shapes (saturating, hump-shaped, sigmoid) are about to get noise wrapped around them.
- Later: model selection weeks will ask “does the data support a Holling III (sigmoid) vs. a Michaelis-Menten (no inflection) shape?” – that question only makes sense because of the vocabulary built today.

Discussion seed (if time remains)

Ask: *besides population growth, where else in your own field might a quantity have a single point of fastest change – a real inflection point – rather than a maximum or minimum?* Good prompt for surfacing that “steepest, not biggest” is a genuinely different (and under-used) question to ask of data.

PART 2 – PODIUM CHEAT SHEET

Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.

0. Open (~2 min)

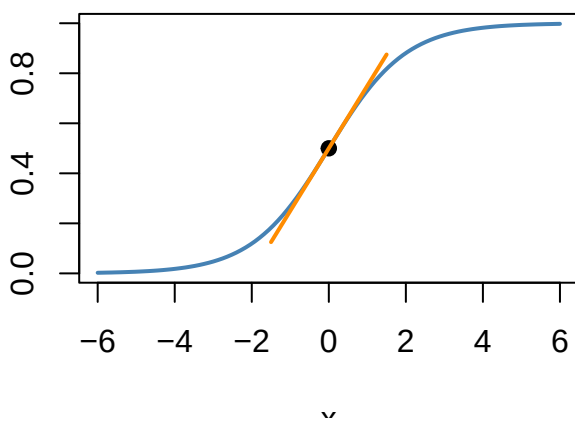
- Monday: toolkit (limits, critical points) via MM and Ricker
- This AM's WOD: limits on 2 new functions + a derivative with a hint
- Today: reveal the hint's answer, then the map

1. WOD4 limits, reveal (~5 min)

- MM, Ricker – already done Monday, just flash it
- Logistic $\frac{1}{1 + e^{-bx}}$: bounded 0 to 1, no new trick needed
- **Holling III** $\frac{ax^2}{b^2 + x^2}$ – do this one on the board:
- $x \rightarrow 0$: numerator wins the “who’s higher order” question this time
 $\rightarrow \approx ax^2/b^2 \rightarrow 0$, quadratically
- $x \rightarrow \infty$: **same degree top and bottom** – new wrinkle! throw out b^2
vs. x^2 separately $\rightarrow ax^2/x^2 = a$
- **Sticking point:** same-degree fractions still use dominance – apply it
inside num. and denom. separately, then simplify

2. Critical points vs. inflection points (~12-14 min)

- Recap: critical point = $f'(x) = 0$ (Ricker's peak, Monday)
- **Logistic derivative** (chain rule twice): $f'(x) = \frac{be^{-bx}}{(1 + e^{-bx})^2}$
- $f'(0) = b/4$ – **not zero!** (matches Bolker Fig. 3.9's own label)



- $x = 0$ = **steepest** point, not flattest $\rightarrow$ **inflection point** ($f'' = 0$, concavity flips), not a critical point
- geometric argument only – concave-up before, concave-down after; no need to grind through f'' symbolically today
- **Payoff:** Monday's sine WOD, day 91 = fastest increase = same feature, unnamed until now. Sigmoid's steepest point and a periodic function's steepest point are the *same calculus idea*.
- **Sticking point:** critical pt = flattest nearby; inflection pt = steepest nearby (for a sigmoid) – easy to blur, keep separate

3. The bestiary map (~8-10 min)

- Table 3.1 (Bolker p.119) – four families, don't derive every row:
- polynomial/piecewise – line, quadratic, hockey stick
- **rational** – MM, Holling III, Holling IV
- **exponential-based** – Ricker, **logistic**
- power-based – power law, von Bertalanffy, Gompertz
- **you've already met about half this table** – today just names it
- **logistic's origin story:** solution to $dn/dt = rn(1 - n/K)$ – per-capita growth declines linearly to zero at carrying capacity K . Mechanistic

view, not just curve-fitting.

- one-liner: Gompertz = another sigmoid, inflection at $1/e$ up not halfway
 - “sigmoid” is a category, not one function

4. Roadmap (~2-3 min)

- Week 3: these shapes become the *mean* of a full stochastic model
- Later: model selection asks “sigmoid or not?” – that question needs today’s vocabulary

If time: discussion seed

- *Where else in your field is there a single point of fastest change – a real inflection point – rather than a max/min?*